

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 1 SET A**SUMMER SEMESTER, 2024-2025****DURATION: 20 MINUTES****FULL MARKS: 15**

SWE 4805: Software Verification and Validation

Write all the answers on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions.

ID:

1.

Consider the following Alloy model shown in Figure 1 and answer the subsequent question.

10
(CO1)
(PO1)

```
1 abstract sig Person {}
2
3 sig Doctor extends Person {
4   patients: set Patient, appointments: set Appointment
5 }
6
7 sig Patient extends Person {
8   ward: lone Ward, appointments: set Appointment
9 }
10
11 sig Nurse extends Person {
12   assignedWards: set Ward
13 }
14
15 sig Ward { doctors: some Doctor, nurses: set Nurse }
16
17 sig Appointment { doctor: one Doctor, patient: one Patient, time: one
18   Time, status: one Status }
19
20 abstract sig Status {}
21 sig Scheduled, Completed, Cancelled extends Status {}
22
23 sig Time {}
24
25 fact appointmentConsistency {
26   all a: Appointment |
27     a.patient in a.doctor.patients
28 }
29
30 fact wardConsistency {
31   all p: Patient | some p.ward => p in p.ward.~nurses.~assignedWards
32 }
33
34 fact noSelfAssignment {
35   all d: Doctor | d not in d.patients
36 }
```

(a) [10 Marks]

(CO1) (PO1)

Describe the above Alloy model in natural language. Your answer should include:

- Purpose of each signature and its role in the system, Meaning of each field, and multiplicity constraint and facts

Solution:

A **hospital management system** that models doctors, patients, nurses, wards, and appointments.

- **Person:** Abstract signature representing all individuals in the hospital system.
- **Doctor:** Extends Person and represents medical professionals. Each doctor:
 - has a set of patients,
 - has a set of appointments.
- **Patient:** Extends Person and represents individuals receiving treatment. Each patient:
 - may belong to at most one ward (`lone Ward`),
 - has a set of appointments.
- **Nurse:** Extends Person and represents supporting medical staff. Each nurse:
 - can be assigned to multiple wards.
- **Ward:** Represents hospital units. Each ward:
 - has one or more doctors (`some Doctor`),
 - has zero or more nurses.
- **Appointment:** Represents a scheduled meeting between a doctor and a patient. Each appointment:
 - is associated with exactly one doctor,
 - is associated with exactly one patient,
 - has exactly one time,
 - has exactly one status (`Scheduled`, `Completed`, or `Cancelled`).
- **Status:** Abstract representation of appointment states.
- **Time:** Represents scheduling time for appointments.

Explanation of Facts:

- **appointmentConsistency:** Ensures that every appointment connects a patient who is actually in the doctor's patient list.
- **wardConsistency:** Ensures consistency between patients, wards, and nurses, meaning ward assignments are properly reflected.
- **noSelfAssignment:** Ensures that a doctor cannot treat themselves as a patient.

*** This is my interpretation, yours can vary ***

2.

5
(CO2)
(PO2)

There are seven common myths about Formal Methods in Software Engineering.

- Any **myths you agree with** (and justify why they are true or partially true)
- Any **myth you disagree with** (and explain why it is incorrect)

Your explanation should include reasoning based on Software Verification and Validation concepts.